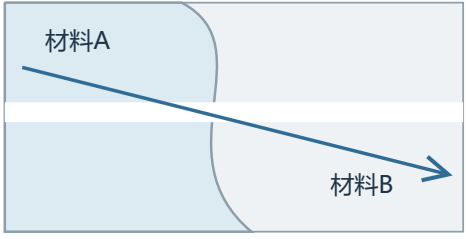
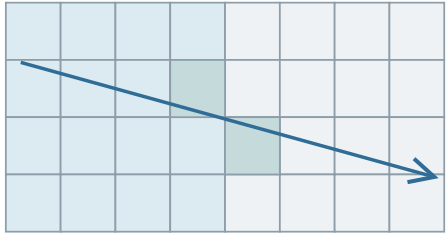
a 曲面与细通道



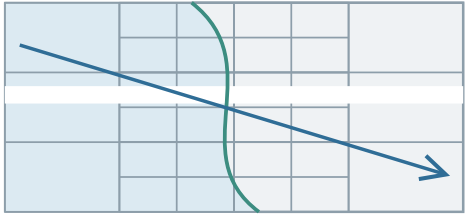
界面位置与通道连通性

b 粗网格材料投影



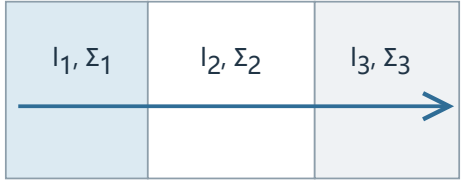
界面单元以混合材料表示

c 界面附近局部离散



细化改变交点及局部源区

d 特征线材料段



$$\tau = \Sigma_1 l_1 + \Sigma_2 l_2 + \Sigma_3 l_3$$